Edison

Literature (High) Results

Objective

Task title: Mechanism-first neighbouring-domain knowledge transfer for OOD scientific discovery

Please read the attached file `EDISON_CLEAN_SHEET_RESEARCH_PROMPT_2026-07-29.md` completely and execute it as the authoritative research brief.

This is a clean-sheet scientific research task, not a manuscript review. Most of our completed cross-database transfer attempts are null, harmful, or too small to matter. Do not defend the current methods and do not restrict yourself to the existing donor-feature injection framework, Random Forest models, composition representations, database pairs or prediction endpoints.

You may search current primary literature and official public repositories globally. The objective is to determine what form of experimentally grounded knowledge is genuinely portable between neighbouring scientific domains, identify suitable donor-recipient data, and design a falsifiable experiment capable of producing a practically meaningful improvement in an OOD region.

Generate new mechanisms and research programmes, not generic transfer-learning suggestions. At least two proposed programmes must be executable with public experimental data and materially different from anything already attempted in the project.

Distinguish verified facts, inference, new hypotheses and items requiring data verification. Do not treat pending experiments as results. Verify all datasets, access links and citations.

The final report must include a ranked set of new hypotheses, a dataset landscape, alternative transfer mechanisms, one preregistration-ready flagship experiment, one OOD scientific-discovery experiment, stop recommendations and a 30-day execution roadmap.

Results

Mechanism-First Neighbouring-Domain Knowledge Transfer for OOD Scientific Discovery

1. Executive Summary

This report addresses a fundamental question: *What form of experimentally grounded knowledge is genuinely portable between neighbouring scientific domains, and how can it be harnessed to achieve practically meaningful out-of-distribution (OOD) improvement?* After surveying 76 papers and systematically evaluating the state-of-the-art, we identify a critical insight: **most cross-database transfer attempts fail because they transfer the**

wrong object (raw features or pretrained weights) **instead of the right object** (mechanistic relationships that govern how properties change under compositional or structural perturbation). We propose six ranked hypotheses, two fully specified experiments executable with public data, five stop recommendations, and a 30-day execution roadmap.

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2. Diagnosis: Why Most Transfer Attempts Are Null or Harmful

2.1 The Pseudo-OOD Problem [VERIFIED FACT]

A rigorous study of OOD generalization in materials science evaluated over 700 OOD tasks across JARVIS, Materials Project, and OQMD. The central finding was that most heuristically defined OOD tests—including leave-one-element-out and leave-one-structure-type-out—do not represent genuinely difficult extrapolation challenges. Analysis of material representation spaces revealed that well-performing "OOD" tasks had test data residing within the training domain, while truly challenging tasks involved data outside the training manifold ([1.1]). Increasing training set size or training time produced only marginal improvements or even degradation for these genuinely hard tasks ([1.2], [1.1]). This means that reported "positive transfer" results may largely reflect interpolation, not true extrapolation.

2.2 Foundation-Model Pretraining Hurts OOD [VERIFIED FACT]

The BOOM benchmark evaluated over 140 model-task combinations. A striking finding is that self-supervised pretraining (masked language modeling) improved in-distribution accuracy by 31–35% but **reduced** OOD binned R^2 by 39–53% for models such as MoLFormer and ChemBERTa ([2.1]). No existing model achieves strong OOD generalization across all tasks: the top-performing model exhibited an average OOD error 3× larger than in-distribution ([2.2]). Electronic-structure properties (HOMO, LUMO, band gap, dipole moment) showed particularly poor OOD performance across all architectures ([2.3]).

2.3 Negative Transfer from Domain Misalignment [VERIFIED FACT]

Multi-task fine-tuning with structurally misaligned auxiliary domains degrades OOD performance. When OMat24 (inorganic materials) or ODAC23 (metal-organic frameworks) were used as auxiliary tasks for QM9 molecular prediction, OOD RMSE increased by 17–22% ([3.1]). Standard fine-tuning causes "representation collapse," where general-purpose chemical priors learned during pretraining are erased, severely degrading OOD generalization ([3.2]).

2.4 Linear Scaling Relations Break Down in Complex Systems [VERIFIED FACT]

The Brønsted–Evans–Polanyi (BEP) and linear scaling relationships (LSR) that historically enabled knowledge transfer across transition-metal catalysts lose predictive power for complex systems: when reaction networks grow larger, when oxide/nitride semiconductors are involved, when coverage effects dominate, and when configurational complexity is high. Uncertainties in activation energies are amplified exponentially through the Arrhenius equation, producing 2–3 orders of magnitude error in rate predictions ([4.1], [4.2], [4.3]).

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3. What *Is* Genuinely Portable?

Based on the evidence, three forms of knowledge survive cross-domain transfer under true OOD conditions:

A. Relative property-change functions ($\Delta x \rightarrow \Delta y$). Rather than predicting absolute property values from new compositions, learning how properties change as a function of compositional or structural differences produces OOD improvements of 2–3× in true-positive rate for both materials and molecules ([5.1], [5.2], [5.3], [5.4]). This "analogical reasoning" approach leverages a deep chemical principle: similar changes in structure imply similar changes in properties ([6.1]).

B. Physically motivated domain-transformation functions. When a known physical law relates source and target quantities, encoding that law as the transfer mechanism (rather than as a feature) achieves order-of-magnitude improvement. Yahagi et al. demonstrated that chemistry-informed domain transformation from DFT to experiment, using statistical ensemble relations and BEP-type conversion functions, achieved test errors one order of magnitude smaller than scratch models, using fewer than 10 experimental calibration points ([7.1], [7.2], [7.3]).

C. Shared potential-energy-surface geometry via small bridging sets. SevenNet-Omni showed that cross-domain training on 15 open datasets spanning molecules, crystals, and surfaces, using domain-bridging sets of only 0.1% of data and selective regularization, achieves state-of-the-art cross-domain accuracy—including adsorption-energy errors below 0.06 eV on metallic surfaces ([8.1], [8.2], [8.3], [8.4]).

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4. Ranked Hypotheses

The following table ranks six new hypotheses for mechanism-first knowledge transfer, each with falsifiable predictions, epistemic status, and expected effect sizes. All are materially different from the existing donor-feature-injection/Random-Forest framework.

Rank	Hypothesis Name	Core Mechanism	Donor→Recipient Domain Pair	Testable Prediction	Epistemic Status (Verified Fact / Inference / New Hypothesis / Requires Data Verification)	Expected Effect Size
1	Analogical Scaling Transfer	Use bilinear transduction to learn relative adsorption-energy change functions under compositional substitution, then transfer BEP-like scaling in difference space rather than absolute-energy space; motivated by strong OC20→OC22 cross-domain gains and analogical OOD prediction results ([9.1], [5.1], [5.2], [5.3])	Metal-surface catalysis (OC20) → oxide electrocatalysis (OC22)	>2x improvement in TPR for identifying OOD oxide catalysts with near-optimal binding energy versus direct absolute-energy prediction baseline ([5.2])	New Hypothesis	Large
2	Chemistry-Informed Residual Bridge	Pretrain a GNN on large donor DFT data, then learn a physics-motivated DFT→experiment correction using chemistry-informed domain transformation grounded in statistical ensembles and source-target quantity relations; based on demonstrated order-of-magnitude gains in catalyst Sim2Real transfer ([7.1], [7.2], [7.3])	DFT band-gap prediction (Materials Project / AFLOW / OQMD class) → sparse experimental band gaps (ICSD-linked experimental set; requires exact dataset verification) ([10.1], [11.1])	MAE <0.3 eV on OOD experimental band gaps with <50 experimental calibration points	Inference	Large
3	Mechanistic Invariant Extraction	Use SHAP-guided causal discovery to identify invariant subgraph→property mechanisms that persist across related chemistries, transferring motifs rather than raw descriptors; motivated by causal-structure extraction in materials and evidence that compositional mechanism, not structure alone, drives hard OOD errors ([1.2])	Experimental molecular solubility domain → sparse polymer membrane permeability domain	R^2 >0.7 on held-out polymer families absent from training	New Hypothesis	Medium–Large
4	Multi-Task Force-Field Regularization for OOD Catalysis	Apply SevenNet-Omni-style shared/task-specific parameterization, selective regularization, and domain-bridging sets so common PES structure learned on metals transfers to unseen catalytic alloys; motivated by successful cross-domain molecules↔crystals↔surfaces transfer and 0.1% bridging sets ([8.1], [8.2], [8.3], [8.4])	Metal adsorption datasets (OC20 plus related open catalyst data) → high-entropy alloy surfaces outside existing training coverage	Force MAE <50 meV/Å on HEA surfaces with zero HEA training data	New Hypothesis; Requires Data Verification	Medium–Large

Rank	Hypothesis Name	Core Mechanism	Donor→Recipient Domain Pair	Testable Prediction	Epistemic Status (Verified Fact / Inference / New Hypothesis / Requires Data Verification)	Expected Effect Size
5	Extreme-Value Data Augmentation Transfer	Mine donor-domain extreme compositions or structures to densify the recipient tail region, exploiting the BOOM result that small fractions of extreme-valued examples materially improve OOD behavior even when generic pretraining fails ([2.4], [2.2], [2.3], [2.1])	AFLOW extreme-composition donor set → JARVIS band-gap recipient model	>30% reduction in tail-region RMSE relative to same architecture without tail augmentation	Inference	Medium
6	Relative Representation Alignment	Train domain-specific encoders, then align them in a shared $\Delta x \rightarrow \Delta y$ space so transferable knowledge is encoded as relational change laws, not absolute feature overlap; motivated by bilinear transduction gains and by evidence that direct pretraining/fine-tuning often collapses OOD representations ([5.1], [5.5], [5.4], [3.2])	Generic neighboring-domain pair with partial mechanistic overlap (e.g., donor computational materials → recipient experimental materials; donor molecular → recipient polymer)	>25% MAE reduction in the true OOD region versus direct feature-transfer baseline	New Hypothesis	Medium

Table: This table ranks six mechanism-first hypotheses for portable scientific knowledge transfer under true OOD conditions. It summarizes each hypothesis, the donor-recipient domain pair, falsifiable prediction, epistemic status, and expected effect size, with citations to the supporting evidence base.

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5. Dataset Landscape

The following table maps the public datasets suitable for donor-recipient pairing in the proposed experiments, with verified access links and role assignments.

Dataset Name	Domain	Size (structures/compounds)	Data Type (DFT/Experimental/Mixed)	Key Properties Available	Access URL	Verified Open Access?	Potential Role
OC20	Heterogeneous catalysis (metals)	~250M single-point calculations	DFT (RPBE)	Adsorption energies, total energies, forces, relaxed structures	https://opencatalystproject.org	Yes	Donor ([9.1], [12.1])
OC22	Oxide electrocatalysis	~62K relaxations / ~9.8M single-point calculations	DFT (RPBE)	Total energies, forces, adsorption energies, oxide surface relaxations	https://opencatalystproject.org	Yes	Recipient ([9.1], [12.1])
AQCat25	Heterogeneous catalysis (spin-aware)	Size requires dataset-page verification	DFT (RPBE+spin)	Adsorption energies, forces, spin-aware catalytic energetics	https://github.com/amazon-science/aqcat25	Yes	Donor/Recipient ([13.1])
Materials Project	Inorganic crystals	~150K compounds	DFT (PBE+U)	Band gap, formation energy, elastic properties, structures	https://materialsproject.org	Yes	Donor ([12.1], [10.1], [11.1])
AFLOW	Inorganic crystals	~3.5M compounds	DFT (PBE)	Band gap, formation energy, thermal properties, elasticity	https://aflow.org	Yes	Donor ([12.1], [10.1], [11.1])

Dataset Name	Domain	Size (structures/compounds)	Data Type (DFT/Experimental/Mixed)	Key Properties Available	Access URL	Verified Open Access?	Potential Role
JARVIS-DFT	Inorganic crystals / 2D materials	~80K compounds	DFT (optB88vdW)	Band gap, formation energy, elastic properties, exfoliation-related properties	https://jarvis.nist.gov	Yes	Both ([10.1], [1.2])
OQMD	Inorganic crystals	~1M compounds	DFT (PBE+U)	Formation energy, phase stability, structures	https://oqmd.org	Yes	Donor ([12.1], [10.1], [11.1], [1.2])
QM9	Small organic molecules	~134K molecules	DFT (B3LYP)	HOMO, LUMO, gap, dipole, thermochemistry	Public via PyG / OGB	Yes	Donor ([2.5], [2.4])
QMugs	Drug-like molecules	~2M conformers	DFT (ωB97X-D)	Quantum properties, conformer-resolved molecular energetics	https://zenodo.org/record/6707529	Yes	Donor ([14.1])
ICSD (exp subset)	Experimental inorganic crystals	~280K entries	Experimental	Crystal structure, experimentally reported properties including band gap for subsets	https://icsd.products.fiz-karlsruhe.de (subscription)	Partially open	Recipient ([11.1])
OMat24	Inorganic bulk materials	Size requires dataset-page verification	DFT (PBE+vdW)	Total energies, forces, bulk-material atomistic data	https://opencatalystproject.org	Yes	Donor ([3.3], [3.1])
Catalysis-Hub	Heterogeneous catalysis	~100K calculations	DFT	Adsorption energies, reaction energies, surface–adsorbate structures	https://www.catalysis-hub.org	Yes	Both ([12.1])
Starrydata2	Thermoelectric materials	~30K experimental entries	Experimental	Seebeck coefficient, electrical conductivity, thermal conductivity, ZT	https://www.starrydata2.org	Yes	Recipient

Table: This table maps public donor and recipient datasets relevant to mechanism-first knowledge transfer across catalysis, inorganic materials, and molecular domains. It is useful for selecting verified data sources and identifying where true cross-domain, OOD transfer experiments can be executed with public resources.

Data verification notes [REQUIRES DATA VERIFICATION]: 1 The experimental band-gap recipient set from ICSD requires subscription access for full records; a partially open alternative is the curated Zhuo et al. 2018 experimental band-gap compilation. 2 AQCat25 dataset size should be confirmed from the repository. 3 Starrydata2 is a fallback experimental recipient for thermoelectric properties if the band-gap recipient set is insufficient.

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6. Alternative Transfer Mechanisms

Six transfer mechanisms have been identified that are fundamentally different from donor-feature injection into Random Forests. Each transfers a different kind of object and has distinct boundary conditions for success vs. failure.

Mechanism Name	What is Transferred	When It Works	When It Fails	Key References	Novelty vs. Prior Project Work
Bilinear Transduction / Analogical Reasoning	Relative property-change functions, mapping (\Delta x \rightarrow \Delta y), using anchor-based analogies rather than absolute property prediction	When compositional or structural substitution patterns are shared across donor and recipient domains, so similar differences imply similar property shifts	When domains lack meaningful analogous substitutions, anchor selection is unstable, or the OOD point cannot be related to any in-support training contrast	Segal et al. 2025; analogical chemistry reasoning summaries ([5.1], [5.2], [5.5], [5.3], [6.1])	Entirely different from donor-feature injection: it learns relational mappings in difference space, not absolute predictions in original feature space
Chemistry-Informed Domain Transformation	Physically motivated correction functions linking source and target quantities, e.g. BEP-type relations, statistical-ensemble mappings, and source-to-target conversion laws	When a known physical or chemical law relates donor and recipient measurements and the correction can be parameterized with limited target data	When the governing law is unknown, strongly regime-dependent, or more nonlinear than the chosen parameterization can capture	Yahagi et al. 2025; catalysis scaling-law context ([7.1], [7.2], [7.3], [4.1])	Encodes domain knowledge as the transfer mechanism itself rather than appending donor outputs as extra features
Domain-Bridging Set + Selective Regularization	Shared potential-energy-surface representations across domains, with small bridging subsets aligning domains and regularization preserving shared structure	When donor and recipient domains share underlying atomic interactions or related PES geometry across molecules, crystals, and surfaces	When bonding character or computational protocol mismatch is too large, so shared parameters become misleading instead of transferable	Kim et al. 2026 / SevenNet-Omni ([8.1], [8.2], [8.3], [8.4], [8.5])	Transfers PES geometry and shared representations, not tabular features; explicitly separates shared and task-specific parameters
Multi-Task Fine-Tuning with Representation Preservation	Pretrained representations preserved during adaptation by auxiliary multi-task losses that prevent collapse of chemically useful priors	When auxiliary tasks have real structural or chemical overlap with the target task and help retain general representations	When auxiliary domains are misaligned; reported examples include OMat24 and ODAC23 hurting QM9 OOD performance, with average RMSE increases around 20% or worse	Zhang et al. 2026; negative-transfer evidence ([3.3], [3.1], [3.2])	Directly targets representation collapse, a failure mode of standard fine-tuning, rather than assuming more donor information is always beneficial
Extreme-Value Augmentation from Neighbor Domain	Tail-distribution samples or extreme-valued examples from a neighboring, data-richer domain	When donor tail compositions or structures overlap the recipient's real OOD region and expose the model to the right extremes	When donor tail regions are themselves sparse, noisy, or chemically irrelevant to the recipient tail of interest	BOOM benchmark / Antoniuk et al. 2026 ([2.4], [2.2], [2.3], [2.1], [2.6])	Transfers data points rather than a pretrained model; specifically addresses OOD tail failure instead of global average performance
Multi-Fidelity Δ-Learning with Denoising	Low-fidelity electronic-structure information plus a learned residual correction to higher-fidelity or experimental targets	When the low-to-high fidelity mapping is smooth, systematic, and sufficiently overlapping in configuration and chemical space	When low-fidelity noise overwhelms signal, overlap is weak, or the baseline method introduces harmful bias, as seen with negative transfer from DFTB-like baselines	Cui et al. 2025; Kim et al. 2024; Liu et al. 2022 ([15.1])	Differs from composition-only transfer by moving the transferable object to electronic-structure-level residuals and denoised fidelity corrections

Table: This table summarizes six transfer mechanisms that are materially different from donor-feature injection and Random Forest baselines. It highlights what each mechanism transfers, the boundary conditions for success versus failure, and why each is a distinct research direction for true OOD scientific discovery.

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7. Experiment A: Preregistration-Ready Flagship — Analogical Scaling Transfer

This experiment tests Hypothesis 1 by applying bilinear transduction to transfer learned Δ(composition)→Δ(adsorption energy) functions from OC20 metal catalysts to OC22 oxide electrocatalysts under a true OOD split.

Component	Specification
Title	Analogical Scaling Transfer: Learning Relative Adsorption-Energy Changes to Predict OOD Oxide Electrocatalysts

Component	Specification
Hypothesis	Transferring learned delta-composition to delta-adsorption-energy functions from metal catalysts to oxide catalysts via bilinear transduction will produce more than 2x improvement in true-positive rate for identifying near-optimal OER oxide catalysts, compared to direct absolute-energy GNN prediction ([9.1], [5.1], [5.2])
Donor Data	OC20 dataset: metal-surface adsorption energies for O-star, OH-star, and OOH-star; about 250 million single-point calculations; open access via Open Catalyst Project ([9.1], [12.1])
Recipient Data	OC22 dataset: oxide-surface adsorption and total-energy data; about 62 thousand relaxations and about 9.8 million single-point calculations; open access via Open Catalyst Project ([9.1], [9.2], [12.1])
OOD Test Definition	Hold out all oxide compositions containing three or more transition-metal species as the true OOD test set; confirm OOD status by UMAP or representation-space analysis showing test samples lie outside the training support region, following materials OOD evaluation guidance ([1.2], [1.1])
Baseline Models	(a) GemNet-OC trained on OC22 only; (b) GemNet-OC pretrained on OC20 then fine-tuned on OC22, replicating the OC20-to-OC22 transfer protocol reported by Tran et al. ([9.1], [9.2])
Experimental Model	Bilinear transduction model trained on OC20 metal composition-substitution pairs to learn delta-composition to delta-Eads functions; apply to OC22 oxide compositions using element-group analogies such as 3d-to-4d-to-5d substitutions and anchor-based relational prediction ([5.1], [5.5], [5.3])
Primary Metric	True Positive Rate at 30 percent extrapolative precision threshold for identifying oxides with absolute deviation of delta-G-O-star from 1.5 eV less than 0.2 eV as near-optimal OER binders; 30 percent extrapolative precision follows prior OOD evaluation practice ([5.6])
Secondary Metrics	MAE on OOD adsorption energies; Spearman rank correlation between predicted and DFT ordering; precision at 20 for top candidates; optionally report calibration or error stratified by composition family ([2.5], [2.3])
Success Criterion	TPR of transduction divided by TPR of baseline b greater than 2.0 and OOD MAE reduction greater than 20 percent versus baseline b
Failure Criterion (Stop)	If TPR improvement is less than 1.2x and MAE improvement is less than 5 percent after hyperparameter sweep and anchor-selection ablation, abandon this mechanism for this donor-recipient pair
Analysis Plan	Pre-register the OOD split before model training; bootstrap 95 percent confidence intervals for all metrics; run ablations on anchor selection, analogy definition, and substitution families; perform post hoc error analysis by oxide composition family and distance from training manifold ([1.2], [1.1], [5.5])
Computational Requirements	About 4 GPU-days for GemNet-OC baselines and about 1 GPU-day for the transduction model; estimated total about 5 to 7 GPU-days depending on sweep breadth
Timeline	Weeks 1 to 2: data preparation, adsorption-target harmonization, OOD split construction; Weeks 2 to 3: baseline training and verification; Weeks 3 to 4: transduction implementation, evaluation, bootstrap analysis, and ablations
Rationale for Mechanism	Prior work shows direct OC20-to-OC22 transfer improves oxide prediction, but most OOD benchmarks overstate difficulty by testing interpolation; analogical transduction is designed specifically for true extrapolation by learning relative changes rather than absolute values ([9.1], [1.1], [5.2], [5.4])

Table: This table defines a preregistration-ready flagship experiment for testing analogical, mechanism-first transfer from metal catalysis to oxide electrocatalysis under a genuinely out-of-distribution split. It specifies datasets, baselines, metrics, stopping rules, and analysis choices grounded in the retrieved evidence.

Why this is materially different from prior work: The OC20→OC22 fine-tuning approach of Tran et al. 2023 achieved ~36% improvement in energy predictions but operates by transferring pretrained weights—an approach susceptible to representation collapse and pseudo-OOD inflation ([9.1], [9.3]). The proposed analogical scaling transfer instead learns *how adsorption energies change under compositional substitution*, predicting in difference space rather than absolute space. This is grounded in the chemical principle that BEP-like scaling relations, while unreliable in absolute form for oxides, may retain predictive power in relative/differential form ([4.1], [5.1]).

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8. Experiment B: OOD Scientific Discovery — Chemistry-Informed Residual Bridge for Band Gap Extrapolation

This experiment tests Hypothesis 2 by predicting experimental band gaps of novel oxide families using a physics-motivated DFT→experiment correction, targeting discovery of ultrawide-bandgap semiconductors.

Component	Specification
Title	Chemistry-Informed Residual Bridge for Predicting Experimental Band Gaps of Novel Oxide Families from DFT
Scientific Discovery Target	Predict experimental band gaps of ternary and quaternary oxide compositions not represented in experimental training, to identify new ultrawide-band-gap semiconductors (>4 eV) for power electronics; this targets true chemistry OOD rather than interpolation ([7.1], [1.1])
Hypothesis	A chemistry-informed domain transformation mapping DFT PBE or PBE+U band gaps to experimental band gaps, using a physically motivated correction function parameterized by d-electron count, electronegativity difference, and a crystal-field-splitting proxy, will achieve OOD MAE <0.3 eV and enable reliable identification of ultrawide-band-gap materials ([7.1], [7.2], [7.3])
Donor Data	Materials Project and/or AFLOW DFT band-gap datasets, with roughly 80K-150K usable inorganic compounds depending on filtering; open access via materialsproject.org and aflow.org; use only entries with converged structures and band-gap values ([12.1], [10.1], [11.1])
Recipient Data	Experimental band gaps compiled from ICSD-linked annotations plus literature-curated datasets; target scale approximately 1,000-3,000 verified compounds, subject to de-duplication and provenance checks; ICSD access is partial/subscription-based, so exact recipient set requires verification before preregistration ([11.1])
OOD Test Definition	Hold out entire chemical families present in donor DFT databases but absent from experimental training, for example quaternary pyrochlores $A_2B_2O_7$ or compounds containing two or more f-block elements; confirm OOD status by representation-space separation rather than heuristic family labels alone ([1.2], [1.1])
Transfer Mechanism	Step 1: train MEGNet or ALIGNN on the full donor DFT set to predict PBE/PBE+U band gap. Step 2: define a chemistry-informed transformation ($E^{exp}_{gap}=f(E^{DFT}_{gap};\theta)$) where ($f$) encodes systematic DFT underestimation as a function of d-electron count, electronegativity difference, and crystal-field proxy. Step 3: calibrate (θ) on fewer than 50 experimental compounds. Step 4: apply the full pipeline to OOD compounds. This follows the demonstrated chemistry-informed domain-transformation logic that achieved order-of-magnitude gains in DFT→experiment transfer in catalysis ([7.1], [7.2], [7.3])

Component	Specification
Baseline	(a) Raw DFT band gaps used directly; (b) GNN trained on experimental data only; (c) standard transfer learning with DFT pretraining then fine-tuning on experiment; optionally include a linear delta-correction baseline and a plain isotonic or affine calibration baseline to test whether chemistry-informed structure matters ([7.1], [2.1], [15.1])
Primary Metric	Mean absolute error on experimental band gaps for held-out OOD chemical families
Discovery Validation	Cross-check the top 20 predicted ultrawide-band-gap candidates against independent higher-fidelity calculations such as HSE06 or GW when available, and against any experimental reports published or curated outside the training set; report confirmation yield and reasons for non-confirmation ([5.6])
Success Criterion	OOD MAE <0.3 eV and at least 50% of the top 20 ultrawide-band-gap predictions supported by HSE06/GW or independent experiment
Failure Criterion	Stop if OOD MAE >0.5 eV, or if the chemistry-informed correction does not outperform a linear delta-E correction, or if results collapse to pseudo-OOD performance only within the training manifold ([1.1])
Timeline	Weeks 1-2: curate and verify the experimental band-gap dataset, harmonize ICSD/literature identities, and define OOD families. Week 3: train DFT baseline models. Week 4: implement and calibrate the chemistry-informed transformation. Weeks 4-5: evaluate on preregistered OOD families and validate top-ranked predictions.

Table: This table specifies Experiment B, a preregistration-ready OOD scientific-discovery study using chemistry-informed transfer from large DFT databases to sparse experimental band-gap data. It defines the donor and recipient datasets, the transfer mechanism, the true-OOD split, validation criteria, and stop rules.

Why this is materially different from prior work: Standard DFT→experiment transfer learning pretrains a GNN on DFT data and fine-tunes on experiment—an approach that the BOOM benchmark shows can *hurt* OOD performance ([2.1]). The chemistry-informed residual bridge instead parameterizes the correction using d-electron count, electronegativity, and crystal-field proxies, following the demonstrated success of Yahagi et al.'s approach where <10 calibration points produced 10× error reduction ([7.1], [7.2]). The key innovation is that the transfer mechanism is the physical correction law itself, not the neural network weights.

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9. Stop Recommendations

Based on the evidence, the following approaches should be stopped or fundamentally restructured:

> *STOP donor-feature injection with Random Forests for cross-database transfer unless a method demonstrates more than 5% improvement in a true OOD region. Current evidence suggests many reported gains are likely interpolation artifacts because most nominal OOD tests in materials are not genuinely outside training support ([1.1], [1.2]).*
>
> *STOP using self-supervised, masked-language-model, or generic foundation-model pretraining as an OOD strategy by default. In the BOOM benchmark, pretraining improved in-distribution results but reduced OOD binned R^2 by 39-53% for models such as ChemBERTa and MoLFormer, showing that better ID performance can coincide with worse extrapolation ([2.2], [2.3], [2.1], [2.6]).*
>
> *STOP naively transferring between structurally misaligned domains without first verifying overlap in configuration space, chemistry, and mechanism. Multi-task fine-tuning with misaligned auxiliary domains such as OMat24 and ODAC23 degraded molecular OOD performance by roughly 17-22% RMSE, demonstrating that indiscriminate transfer can be actively harmful ([3.1], [3.2]).*
>
> *STOP treating random-split or in-distribution evaluation as evidence of transfer success. Pre-register the OOD split, confirm that test points lie outside the training manifold with representation-space diagnostics, and only then claim OOD transfer benefit ([1.1], [1.2]).*
>
> *PROCEED CAUTIOUSLY with low-fidelity donor data. Multi-fidelity transfer can fail when donor noise is high; DFTB-level baselines were specifically identified as a source of negative transfer, so donor-noise auditing and overlap checks should be mandatory before adopting any multi-fidelity scheme ([15.1]).*

Blockquote: This blockquote lists five concrete stop rules for the programme, based on evidence about pseudo-OOD evaluation, harmful pretraining, domain misalignment, and noisy low-fidelity donors. It is useful for deciding which existing directions to terminate before investing more execution time.

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10. 30-Day Execution Roadmap

Day Range	Activity	Deliverable	Go/No-Go Gate
Days 1-3	Dataset acquisition and verification. Download OC20/OC22 subsets; verify AFLOW and Materials Project band-gap access; compile the experimental band-gap recipient set from ICSD and literature; confirm fallback access to Starrydata2 for an alternative recipient task ([9.1], [12.1], [10.1], [11.1])	Verified data manifest with record counts, property distributions, provenance, deduplication status, and confirmed access routes	If the experimental band-gap set contains fewer than 500 usable compounds after harmonization, pivot to Starrydata2 thermoelectric data as the recipient task
Days 3-5	OOD split construction. Define holdout families for Experiment A (ternary-plus oxide catalysts) and Experiment B (quaternary oxide band gaps); validate with UMAP/PCA or equivalent representation-space analysis that the test sets lie outside the training support ([1.2], [1.1])	Pre-registered OOD split files plus UMAP/PCA visualizations demonstrating separation	If fewer than 10% of test points are genuinely OOD by representation analysis, redefine the split using stricter family and manifold-distance criteria
Days 5-10	Baseline model training. Train GemNet-OC baselines for Experiment A (OC22-only and OC20→OC22 fine-tune); train ALIGNN or MEGNet baselines for Experiment B (DFT-only and experimental-only) ([9.1], [9.2], [1.2])	Baseline performance tables with in-distribution and OOD metrics, including TPR, MAE, and rank correlation	If the OC22-only baseline already achieves TPR > 0.6 on the OOD set, the challenge is too easy; tighten the split before proceeding

Day Range	Activity	Deliverable	Go/No-Go Gate
Days 10-15	Implement core transfer mechanisms. Build the bilinear transduction model for Experiment A and the chemistry-informed domain-transformation model for Experiment B ([7.1], [7.2], [5.1], [5.5])	Working, version-controlled code for both proposed transfer approaches	N/A during implementation phase
Days 15-20	Run main experiments and hyperparameter sweeps. Execute Experiment A with anchor-selection ablations; execute Experiment B with correction-function ablations and calibration-set sensitivity checks ([5.1], [5.2], [5.3])	Raw results tables for all model variants and sweep settings	If both experiments show less than 5% improvement over the best baseline, invoke the stop protocol and pivot to Hypothesis 5: extreme-value augmentation ([2.4], [2.2])
Days 20-25	Statistical analysis and validation. Bootstrap 95% confidence intervals; for Experiment B, cross-check top-20 predictions against HSE06 or literature where available; perform error analysis by composition family and OOD distance ([1.2], [1.1], [5.6])	Complete statistical analysis package with confidence intervals, validation notes, and composition-stratified error analysis	If confidence intervals overlap all baselines and no candidate-validation signal emerges, downgrade claims to null or inconclusive and do not proceed to preregistration
Days 25-28	Ablation studies and mechanism diagnosis. Ablate anchor-selection method, analogy definition, correction-function form, and fraction of bridging data; diagnose which compositional families benefit from or are harmed by transfer ([8.1], [8.4], [5.3], [15.1])	Ablation results identifying the minimum viable mechanism and the boundary conditions for positive versus negative transfer	If gains disappear under minimal ablations or are confined to pseudo-OOD subsets, classify the mechanism as non-robust and stop further scaling
Days 28-30	Report writing and preregistration. Compile the final technical report; if results are positive and robust, draft a preregistration for prospective validation; otherwise produce a pivot or stop recommendation ([1.1], [2.1], [3.1])	Final report plus preregistration document if warranted, or a documented pivot recommendation	Proceed to preregistration only if at least one experiment clears its success criterion and survives ablation and CI checks

Table: This table lays out a concrete 30-day plan for executing the two proposed transfer programmes, including dataset verification, true-OOD split construction, model development, validation, and decision gates. It is useful because it turns the research brief into operational milestones with explicit stop and pivot criteria.

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11. Key Distinctions: Fact, Inference, Hypothesis, Requires Verification

Claim	Status
Most materials OOD tests are pseudo-OOD (interpolation)	**Verified Fact** ([1.1])
Foundation-model pretraining reduces OOD R^2 by 39–53%	**Verified Fact** ([2.1])
Misaligned multi-task fine-tuning causes 17–22% RMSE increase	**Verified Fact** ([3.1])
OC20→OC22 cross-training yields ~36% energy improvement	**Verified Fact** ([9.1])
Bilinear transduction achieves 3× TPR improvement for OOD materials	**Verified Fact** ([5.2])
Chemistry-informed domain transformation achieves 10× error reduction	**Verified Fact** ([7.1], [7.2])
Domain-bridging sets of 0.1% data suffice for cross-domain alignment	**Verified Fact** ([8.2], [8.4])
Analogical scaling transfer will work for metal→oxide catalysis	**New Hypothesis** (untested combination)
Chemistry-informed bridge will achieve <0.3 eV OOD band-gap MAE	**Inference** (extrapolating from catalysis demonstration)

Claim	Status
ICSD experimental band gap set has >500 usable compounds	**Requires Data Verification**
AQCat25 dataset is fully compatible with OC20 adsorbate definitions	**Requires Data Verification**

12. Conclusion

The central finding of this analysis is that the object of transfer must change. Transferring pretrained weights or donor features into recipient models has a high failure rate because (a) most apparent successes reflect pseudo-OOD interpolation, (b) generic pretraining actively harms OOD extrapolation, and (c) domain misalignment causes negative transfer. The evidence points to three genuinely portable forms of knowledge: relative property-change functions, physically motivated correction laws, and shared PES geometry via minimal bridging sets. The two proposed experiments—Analogical Scaling Transfer (Experiment A) and Chemistry-Informed Residual Bridge (Experiment B)—are designed to test whether these mechanisms can produce practically meaningful OOD improvement in catalysis and semiconductor discovery, using entirely public data and pre-registered evaluation protocols with explicit stop criteria.

References

[1.1] Probing out-of-distribution generalization in machine learning for materials. Kangming Li, Andre Niyongabo Rubungo, Xiangyun Lei, Daniel Persaud, Kamal Choudhary, Brian DeCost, Adji Bousso Dieng, Jason Hattrick-Simpers. Communications Materials (2025). https://doi.org/10.1038/s43246-024-00731-w Context: "This paper from Li et al. (2025) examines out-of-distribution (OOD) generalization in materials science ML across over 700 OOD tasks. The key finding is that most heuristically-defined OOD tests do not represent genuinely difficult challenges—they primarily test interpolation rather than true extrapolation. ML models, including simple tree ensembles, show robust performance across most tasks featuring new chemistry or structural groups absent from training data. However, analysis of material representation spaces reveals that well-performing tasks have test data residing within the training domain, while poorly-performing tasks contain marginal or out-of-domain test data. Importantly, traditional scaling laws do not apply to challenging OOD tasks; increasing training set size or training time produces only marginal improvements or even degradation in generalization performance. The authors conclude that perceived generalizability benefits are likely overstated due to domain misidentification that conflates interpolation with true extrapolation."

[1.2] Probing out-of-distribution generalization in machine learning for materials. Kangming Li, Andre Niyongabo Rubungo, Xiangyun Lei, Daniel Persaud, Kamal Choudhary, Brian DeCost, Adji Bousso Dieng, Jason Hattrick-Simpers. Communications Materials (2025). https://doi.org/10.1038/s43246-024-00731-w Context: "The Li et al. study evaluated out-of-distribution (OOD) generalization for formation energy prediction across three materials databases (JARVIS, Materials Project, OQMD) using multiple ML architectures including random forests, XGBoost, neural networks, graph neural networks (ALIGNN), and large language models (LLM-Prop). For leave-one-element-out tasks, the ALIGNN model achieved R2 scores above 0.95 in 85% of cases, while XGBoost achieved this in 68% of cases. However, performance degraded significantly for nonmetal elements like H, F, and O, where systematic biases emerged. Analysis using SHAP values revealed that poor OOD performance for H, F, and O was primarily due to chemical/compositional differences rather than structural factors. Different models showed varying strengths across difficulty levels—RF performed best on the most challenging tasks despite lower in-distribution performance, while ALIGNN excelled on less demanding tasks. The study demonstrated that while broad OOD generalization across much of the periodic table is achievable, systematic biases and overestimations occur for specific chemical systems."

[2.1] BOOM: Benchmarking Out-Of-distribution Molecular Property Predictions of Machine Learning Models. Evan R. Antoniuk, Shehtab Zaman, Tal Ben-Nun, Peggy Li, James Diffenderfer, Busra Demirci, Obadiah Smolenski, Tim Hsu, A. Hiszpanski, Kenneth Chiu, B. Kailkhura, B. V. Essen. ArXiv (2026). https://doi.org/10.48550/arxiv.2505.01912 Context: "The excerpt demonstrates negative transfer in chemical property prediction where pretraining hurts out-of-distribution (OOD) performance. While language model pretraining (MLM/PLM) significantly improves in-distribution (ID) performance for ChemBERTa (31%), MoLFormer (35%), and Regression Transformer (12%), it provides negligible or negative benefits for OOD generalization. Notably, binned OOD R^2 scores decreased substantially—by 53% for MoLFormer and 39% for ChemBERTa. The research shows that supervised property pretraining only improves OOD performance when sufficiently correlated with the downstream task (Pearson correlation < 0.35). The authors hypothesize that current pretraining objectives fail to convey relevant chemical information for extrapolation to new chemical spaces. Additionally, large models like MoLFormer overfit to ID data despite strong ID performance, achieving poor OOD generalization. Tasks involving electronic structure properties (HOMO, LUMO, Gap, Dipole Moment) show particularly poor OOD performance across all models, suggesting that explicit electronic structure information in molecular representations is lacking."

[2.2] BOOM: Benchmarking Out-Of-distribution Molecular Property Predictions of Machine Learning Models. Evan R. Antoniuk, Shehtab Zaman, Tal Ben-Nun, Peggy Li, James Diffenderfer, Busra Demirci, Obadiah Smolenski, Tim Hsu, A. Hiszpanski, Kenneth Chiu, B. Kailkhura, B. V. Essen. ArXiv (2026). https://doi.org/10.48550/arxiv.2505.01912 Context: "The BOOM benchmark evaluated over 140 combinations of models and property prediction tasks across 10 molecular property datasets to assess out-of-distribution

(OOD) generalization. A key finding is that no existing ML models achieve strong OOD generalization across all tasks, with the top-performing model showing an average OOD error 3x larger than in-distribution error. Deep learning models with high inductive bias perform reasonably on OOD tasks with simple, specific properties, but chemical foundation models with transfer and in-context learning capabilities do not demonstrate strong OOD extrapolation abilities despite promising performance in limited training data scenarios. The benchmark reveals that OOD performance is significantly impacted by data generation, pre-training strategies, hyperparameter optimization, model architecture, and molecular representation choices."

[2.3] BOOM: Benchmarking Out-Of-distribution Molecular Property Predictions of Machine Learning Models. Evan R. Antoniuk, Shehtab Zaman, Tal Ben-Nun, Peggy Li, James Diffenderfer, Busra Demirci, Obadiah Smolenski, Tim Hsu, A. Hiszpanski, Kenneth Chiu, B. Kailkhura, B. V. Essen. ArXiv (2026). https://doi.org/10.48550/arxiv.2505.01912 Context: "The BOOM benchmark reveals that ML models for molecular property prediction show substantial OOD performance degradation across most tasks. Key findings include: (1) All models exhibit S-shaped parity plots on OOD data, indicating they cluster OOD samples but fail to extrapolate beyond training distributions; (2) OOD performance is task-dependent—some properties (HoF, Density, ZPVE, Cv) achieve near ID-level performance, while electronic structure properties (HOMO, LUMO, Gap, Dipole Moment) show poor OOD generalization across all models; (3) Large models like MoLFormer achieve state-of-the-art ID performance but poor OOD performance, suggesting overfitting; (4) Pre-training on large molecular datasets improves ID performance (31-35% for foundation models) but provides negligible or negative OOD improvements, with Binned OOD R^2 decreasing 39-53%; (5) Supervised pre-training helps OOD performance only when pretraining and finetuning tasks are weakly correlated ($r<0.35$). The excerpt focuses on molecular properties rather than materials science specifically, and provides task-level performance variations rather than precise extrapolation vs. interpolation comparisons."

[2.4] BOOM: Benchmarking Out-Of-distribution Molecular Property Predictions of Machine Learning Models. Evan R. Antoniuk, Shehtab Zaman, Tal Ben-Nun, Peggy Li, James Diffenderfer, Busra Demirci, Obadiah Smolenski, Tim Hsu, A. Hiszpanski, Kenneth Chiu, B. Kailkhura, B. V. Essen. ArXiv (2026). https://doi.org/10.48550/arxiv.2505.01912 Context: "The BOOM benchmark evaluated 12 ML model architectures on out-of-distribution (OOD) molecular property prediction tasks using QM9 datasets. Key findings show that no single model achieves strong OOD performance across all tasks, indicating significant generalization challenges. Models exhibit substantial performance degradation on OOD splits compared to in-distribution (ID) data. While 3D GNN models with equivariant symmetries substantially outperform SMILES-based transformer models, hyperparameter optimization provided limited improvements—achieving only 23-60% RMSE reductions for simpler properties. Graph-based representations substantially outperform sequence-based models for OOD tasks. Data augmentation with extreme-valued molecules showed consistent improvements across 7 of 8 properties, even with small fractions of augmented training data (4%), suggesting training data composition critically impacts OOD generalization. ModernBERT improved over other transformer models by 58-78% for certain properties. The study emphasizes that current molecular property prediction models struggle significantly with true extrapolation beyond training regimes and highlights OOD generalization as a critical frontier for future model development."

[2.5] BOOM: Benchmarking Out-Of-distribution Molecular Property Predictions of Machine Learning Models. Evan R. Antoniuk, Shehtab Zaman, Tal Ben-Nun, Peggy Li, James Diffenderfer, Busra Demirci, Obadiah Smolenski, Tim Hsu, A. Hiszpanski, Kenneth Chiu, B. Kailkhura, B. V. Essen. ArXiv (2026). https://doi.org/10.48550/arxiv.2505.01912 Context: "The BOOM benchmark evaluates out-of-distribution (OOD) generalization in molecular property prediction across multiple machine learning models. Key findings show that no single model clearly outperforms others on both in-distribution (ID) and OOD tasks. The Equivariant Transformer (ET) achieves the best overall ID performance on 4 out of 10 tasks, while MACE performs best on OOD prediction for 5 out of 10 tasks. A critical finding is that models performing poorly on OOD data exhibit characteristic S-shaped parity plots, indicating they can cluster OOD samples but fail to extrapolate beyond the training data distribution. The benchmark evaluates RMSE prediction performance as the primary metric for measuring OOD generalization, along with binned R2 values on distribution tails. The study focuses on property space OOD generalization for small molecules, distinguishing it from prior work on inorganic materials that focused on input space distribution shifts."

[2.6] BOOM: Benchmarking Out-Of-distribution Molecular Property Predictions of Machine Learning Models. Evan R. Antoniuk, Shehtab Zaman, Tal Ben-Nun, Peggy Li, James Diffenderfer, Busra Demirci, Obadiah Smolenski, Tim Hsu, A. Hiszpanski, Kenneth Chiu, B. Kailkhura, B. V. Essen. ArXiv (2026). https://doi.org/10.48550/arxiv.2505.01912 Context: "The excerpt discusses instances where pretraining and model optimization hurt out-of-distribution (OOD) performance in molecular property prediction. Specifically, it notes that commonly employed molecular pretraining strategies like masked language modeling often result in decreased OOD performance. The authors identify overfitting to in-distribution (ID) molecules as a key factor that hurts OOD generalization. Additionally, they observe that default hyperparameters are typically optimized to maximize ID performance, which can negatively impact OOD generalization capabilities. While hyperparameter optimization specifically targeted at OOD performance showed some improvements (up to 60% reduction in RMSE for certain properties like density, heat of formation, and ZPVE), the models still struggle to generalize significantly beyond the training regime. These findings suggest that model design choices optimized for ID performance may inherently conflict with OOD generalization."

[3.1] Multi-Task Fine-Tuning Enables Robust Out-of-Distribution Generalization in Atomistic Models. Chengqian Zhang, Duoduo Zhang, Anyang Peng, Mingyu Guo, Yuzhi Zhang, Lei Wang, Guolin Ke, Linfeng Zhang, Tiejun Li, Han Wang. ArXiv (2026). https://doi.org/10.48550/arxiv.2601.08486 Context: "The excerpt demonstrates negative transfer effects in multi-task fine-tuning for atomistic models. Negative transfer occurs when auxiliary domains diverge substantially from the target task. Specifically, when using domain-misaligned auxiliary force-field tasks (OMat24 for inorganic materials, OC20M for heterogeneous catalysis, ODAC23 for metal-organic frameworks) for QM9 molecular property prediction, out-of-distribution (OOD) performance degrades markedly, with average RMSE increases of 20.3%, 17.8%, and 22.4% respectively. Notably, performance with OMat24 and ODAC23 was even inferior to standard fine-tuning. However, OC20M showed slight improvement due to partial structural overlap between catalytic adsorbates and small organic molecules. The key finding is that knowledge transfer is not universally beneficial—indiscriminate transfer across mismatched domains degrades downstream model performance, while limited structural similarity between auxiliary and target domains can still provide benefits."

[3.2] Multi-Task Fine-Tuning Enables Robust Out-of-Distribution Generalization in Atomistic Models. Chengqian Zhang, Duoduo Zhang, Anyang Peng, Mingyu Guo, Yuzhi Zhang, Lei Wang, Guolin Ke, Linfeng Zhang, Tiejun Li, Han Wang. ArXiv (2026). https://doi.org/10.48550/arxiv.2601.08486 Context: "The excerpt discusses representation collapse as a mechanism through which fine-tuning can hurt out-of-distribution (OOD) generalization in atomistic models. The authors identify that standard fine-tuning (FT) of pretrained models leads to distortion of general-purpose prior knowledge during downstream adaptation, causing severe OOD generalization degradation. This representation collapse phenomenon is observed across both molecular and materials systems, appearing to be a broadly occurring failure mode rather than model-specific. The paper proposes that the performance gap between in-distribution (ID) and OOD generalization is closely associated with this representation collapse in the learned representation space. However, the excerpt does not specifically address multi-fidelity transfer learning or provide detailed conditions characterizing when negative transfer occurs in chemistry and materials science ML beyond identifying representation collapse as a key mechanism."

[3.3] Multi-Task Fine-Tuning Enables Robust Out-of-Distribution Generalization in Atomistic Models. Chengqian Zhang, Duoduo Zhang, Anyang Peng, Mingyu Guo, Yuzhi Zhang, Lei Wang, Guolin Ke, Linfeng Zhang, Tiejun Li, Han Wang. ArXiv (2026). https://doi.org/10.48550/arxiv.2601.08486 Context: "The excerpt describes negative transfer effects in atomistic machine learning models. Standard fine-tuning of pretrained models causes representation collapse, where chemical and structural priors learned during pretraining are erased, severely degrading out-of-distribution (OOD) generalization. The OOD performance becomes substantially worse than in-distribution performance. In the BOOM benchmark for organic molecules, top-performing models show OOD errors up to 3 times higher than ID performance when property values lie significantly outside the training range. For both organic molecules and inorganic crystalline materials, performance degradation occurs particularly under property-based and structure-based OOD settings, especially with limited training data. Linear probing avoids some OOD degradation but significantly hurts in-distribution generalizability. The paper proposes multi-task fine-tuning (MFT) as a solution that jointly optimizes downstream tasks while preserving pretrained knowledge."

[4.1] Dimensionality reduction of complex reaction networks in heterogeneous catalysis: From linear-scalingrelationships to statistical learning techniques. Sergio Pablo-García, Rodrigo García-Muelas, Albert Sabadell-Rendón, Núria López. WIREs Computational Molecular Science (2021). https://doi.org/10.1002/wcms.1540 Context: "Linear scaling relationships (LSR) and Brønsted–Evans–Polanyi (BEP) relations enable knowledge transfer by establishing universal correlations between thermochemistry and kinetics. The universality arises from the locality of chemical bonds and symmetry in adsorbate-surface interactions, allowing transition state energies to depend on initial or final states with universal factors of 1.0, independent of energy references. This enables volcano plots and multi-scale modeling across different catalysts. However, LSR break down in complex systems: for oxidation processes, single descriptors prove insufficient and multivariable approaches are needed; extensions to oxides and nitrides require refinement due to semiconductor properties; coverage effects and active site complexity can necessitate iterative parameter tuning to match experiments. The excerpt indicates that errors from LSR-derived parameters get magnified in exponential terms (Arrhenius equation), particularly affecting selectivity predictions. Alternative approaches mentioned include machine learning techniques for more accurate descriptors and Bayesian statistics for kinetic parameter estimation."

[4.2] Dimensionality reduction of complex reaction networks in heterogeneous catalysis: From linear-scalingrelationships to statistical learning techniques. Sergio Pablo-García, Rodrigo García-Muelas, Albert Sabadell-Rendón, Núria López. WIREs Computational Molecular Science (2021). https://doi.org/10.1002/wcms.1540 Context: "The excerpt discusses linear scaling relationships (LSR) in heterogeneous catalysis that create constraints in chemical space defining volcano plots for activity and selectivity. However, the document identifies critical limitations of these relationships: in emerging catalytic processes with increased intermediates, configurational effects, lateral interactions, and complex low-symmetry materials, LSR lose predictive power due to error accumulation. The excerpt indicates that statistical learning and machine learning techniques are emerging as alternative approaches to address complexity and reduce dimensionality of reaction networks. The authors emphasize the importance of developing AI algorithms with interpretability to provide scientific understanding. However, the excerpt does not provide specific mechanisms of knowledge transfer or detailed explanations of when and why LSR break down, nor does it comprehensively detail alternative approaches beyond mentioning statistical learning techniques generally."

[4.3] Dimensionality reduction of complex reaction networks in heterogeneous catalysis: From linear-scalingrelationships to statistical learning techniques. Sergio Pablo-García, Rodrigo García-Muelas, Albert Sabadell-Rendón, Núria López. WIREs Computational Molecular Science (2021). https://doi.org/10.1002/wcms.1540 Context: "The excerpt discusses linear scaling relationships (LSR) that appear for energies of intermediates and transition states in catalytic systems, which arise from dependencies in chemical bond character. However, the document emphasizes significant limitations: as reaction networks grow larger, LSR predictions become increasingly uncertain, particularly for selectivity. Uncertainties in activation energies are magnified exponentially through the Arrhenius equation, resulting in errors of 2-3 orders of magnitude. The text notes that traditional microkinetic models using LSR are limited and cannot accurately predict reaction rates independently. As an alternative approach, the excerpt highlights statistical learning techniques as powerful tools for predicting and inspecting chemical system behavior. The document also mentions automated approaches using graph theory, automation frameworks, and machine learning techniques to address the shortcomings of conventional LSR-based predictions in complex catalytic systems."

[5.1] Known Unknowns: Out-of-Distribution Property Prediction in Materials and Molecules. Nofit Segal, Aviv Netanyahu, Kevin P. Greenman, Pulkit Agrawal, Rafael Gómez-Bombarelli. ArXiv (2025). https://doi.org/10.48550/arxiv.2502.05970 Context: "The transductive approach uses Bilinear Transduction for out-of-distribution property prediction by learning how property values change based on compositional or structural differences rather than predicting values directly. The method is built on the chemical principle that similar materials have similar properties and similar changes in structure imply similar changes in properties. During training, a bilinear predictor learns from difference vectors ($\Delta x = x_i - x_j$) and anchor points, predicting property y_i from material x_j and their difference. At test time, for an out-of-distribution sample, the method selects the best in-distribution anchor point by minimizing the distance between the test-anchor difference and differences in the training distribution. It then predicts the property using the learned relationship between structural/compositional changes and property changes. For solids, analogies involve elemental substitutions (e.g., replacing one f-block element); for molecules, analogies involve structural similarities measured by Maximum Common Structure. This enables zero-shot extrapolation to higher property ranges by leveraging learned patterns of how property changes correlate with compositional or structural variations."

[5.2] Known Unknowns: Out-of-Distribution Property Prediction in Materials and Molecules. Nofit Segal, Aviv Netanyahu, Kevin P. Greenman, Pulkit Agrawal, Rafael Gómez-Bombarelli. ArXiv (2025). https://doi.org/10.48550/arxiv.2502.05970 Context: "Segal et al. propose a transductive approach using the Bilinear Transduction method for out-of-distribution property prediction. Rather than predicting properties directly from new materials, the method reparameterizes the problem to make predictions based on analogical relations between a known training example and a test candidate. Specifically, predictions are conditioned on the difference in representation space between a training material/molecule and the new sample. This leverages analogical input-target relations in both training and test sets to enable zero-shot extrapolation beyond the training data's property value range. By learning how properties change as a function of material differences rather than predicting absolute values from new materials, the approach achieves significant improvements: 3x and 2.5x improvements in True Positive Rate for materials and molecules respectively, and 2x and 1.5x improvements in precision compared to non-transductive baselines."

[5.3] Known Unknowns: Out-of-Distribution Property Prediction in Materials and Molecules. Nofit Segal, Aviv Netanyahu, Kevin P. Greenman, Pulkit Agrawal, Rafael Gómez-Bombarelli. ArXiv (2025). https://doi.org/10.48550/arxiv.2502.05970 Context: "Analogical reasoning for molecular and materials property prediction uses bilinear transduction to enable out-of-distribution (OOD) extrapolation. The method works by identifying in-distribution anchor points that form analogies with OOD targets. Specifically, the relational difference between an OOD point and its selected anchor mirrors the difference between a training anchor-target pair. Two similarity modes are employed: one where training and test pairs are analogous in structure, and another where anchors are similar to each other and targets are similar to each other. During anchor selection, the model can either choose anchors very similar to the OOD target (leveraging feature space similarity) or select anchors that differ from the OOD target but maintain similar relationships to their training counterparts. This approach achieves 3x improvement in true positive rate for OOD detection, enabling prediction of materials and molecules with outstanding property values by leveraging relative representations rather than absolute feature values."

[5.4] Known Unknowns: Out-of-Distribution Property Prediction in Materials and Molecules. Nofit Segal, Aviv Netanyahu, Kevin P. Greenman, Pulkit Agrawal, Rafael Gómez-Bombarelli. ArXiv (2025). https://doi.org/10.48550/arxiv.2502.05970 Context: "The excerpt discusses using analogical reasoning for out-of-distribution (OOD) property prediction through the Bilinear Transduction method. Rather than predicting absolute property values from new materials directly, the approach reparameterizes the problem by making predictions based on analogies to known training examples. Specifically, predictions are made by comparing the representation space difference between a new candidate material/molecule and a chosen training example, then learning how property values change as a function of these differences. This relational approach leverages analogical input-target relations between training and test sets, enabling generalization beyond the training property value range. The method achieves significant improvements in OOD prediction accuracy—improving True Positive Rate by 3x for materials and 2.5x for molecules, and precision by 2x and 1.5x respectively compared to non-transductive baselines. This represents a shift from learning absolute property prediction functions to learning relative property change functions."

[5.5] Known Unknowns: Out-of-Distribution Property Prediction in Materials and Molecules. Nofit Segal, Aviv Netanyahu, Kevin P. Greenman, Pulkit Agrawal, Rafael Gómez-Bombarelli. ArXiv (2025). https://doi.org/10.48550/arxiv.2502.05970 Context: "The excerpt describes Bilinear Transduction, a transductive regression method for out-of-distribution generalization. It works by reparameterizing predictions using differences between data points rather than absolute values. The predictor uses a bilinear function hθ(Δx, x) = fθ(Δx)gθ(x) that combines non-linear embeddings of data point differences (Δx) and anchor points (x). For test points, the method selects an anchor point from the training data that minimizes the distance to the test point's difference vector. This converts the out-of-distribution problem into a within-support problem. The authors adapted Bilinear Transduction for their setting where the output Y is out-of-distribution rather than the input, though they note theoretical guarantees may not fully apply in this case."

[5.6] Known Unknowns: Out-of-Distribution Property Prediction in Materials and Molecules. Nofit Segal, Aviv Netanyahu, Kevin P. Greenman, Pulkit Agrawal, Rafael Gómez-Bombarelli. ArXiv (2025). https://doi.org/10.48550/arxiv.2502.05970 Context: "The paper by Segal et al. demonstrates a bilinear transduction approach for out-of-distribution (OOD) property prediction in both solid materials and molecules. The method is evaluated on multiple benchmarks including AFLOW, Matbench, and Materials Project for solids, and MoleculeNet for molecules. The bilinear transduction approach consistently outperforms baseline methods (Ridge Regression, MODNet, CrabNet for solids; Chemprop, Random Forest, MLP for molecules) in terms of mean average error (MAE) for OOD predictions. Notably, the method demonstrates superior extrapolation capabilities beyond the training distribution support threshold. The paper also introduces a 30% extrapolative precision metric to evaluate the model's ability to identify high-performance extremes—the top 30% of test samples with highest property values. The bilinear transduction method produces predictions closer to the ground truth OOD distribution, particularly evident in band gap and bulk modulus predictions. However, the excerpt does not explicitly describe the analogical input-target relation mechanism or the detailed mechanics of how zero-shot extrapolation to higher property ranges is achieved."

[6.1] Automating Reasoning in Chemical Science and Engineering. Tyler Josephson. ChemRxiv (2025). https://doi.org/10.26434/chemrxiv-2025-q9bb1 Context: "The excerpt explains analogical reasoning in chemistry as a form of inductive reasoning where properties of unknown substances are predicted based on similarities to well-characterized ones. For example, properties of novel polyfluorinated compounds can be estimated by analogy to known compounds like PFOA. The document describes a computational approach called "transduction" for analogical reasoning in molecular property prediction. Rather than fitting a general model (induction), transduction learns contrasts between input/output pairs in training data. During prediction for an unknown molecule, the algorithm finds the most similar input/output pair in the training set and uses the learned contrast to make predictions. This approach implicitly uses relational representations by comparing unknown molecules to similar known examples, potentially enabling better out-of-distribution extrapolation by reasoning about relative differences rather than absolute values."

[7.1] Transfer learning from first-principles calculations to experiments with chemistry-informed domain transformation. Yuta Yahagi, Kiichi Obuchi, Fumihiko Kosaka, Kota Matsui. Machine Learning: Science and Technology (2025). https://doi.org/10.1088/2632-2153/adcdc0 Context: "This paper presents a Sim2Real transfer learning framework that bridges first-principles calculations (DFT) and experimental data for materials science. The key mechanism enabling knowledge transfer is chemistry-informed domain transformation, which leverages two forms of prior knowledge: (1) statistical ensemble concepts and (2) relationships between computational and experimental quantities. The method transforms computational data from simulation space into experimental data space while integrating these qualitatively different domains through underlying physics and chemistry principles. The demonstrated mechanism for portable knowledge is the use of theoretically-grounded domain transformations from theoretical chemistry. The approach achieved positive transfer, reaching high accuracy with fewer than ten experimental data points, performing one order of magnitude better than scratch models requiring over 100 target data points, indicating that chemistry-informed knowledge effectively enables out-of-distribution improvements across the DFT-to-experiment boundary."

[7.2] Transfer learning from first-principles calculations to experiments with chemistry-informed domain transformation. Yuta Yahagi, Kiichi Obuchi, Fumihiko Kosaka, Kota Matsui. Machine Learning: Science and Technology (2025). https://doi.org/10.1088/2632-2153/adcdc0 Context: "This paper demonstrates transfer learning between first-principles computational data and experimental data for catalyst activity prediction. The key mechanism enabling knowledge transfer is chemistry-informed domain transformation using knowledge-based hypotheses about statistical ensembles and conversion functions that reflect underlying chemical laws. The method achieved significant improvements: test losses were reduced by one order of magnitude compared to models trained only on target data. The transferred model reached nearly optimal parameters using only about 10 target data points for domain transformation. The authors note the importance of designing conversion functions based on chemical principles, specifically the Brønsted–Evans–Polanyi (BEP) relation linking activation energy to adsorption energy. However, they acknowledge the approach has limitations when BEP relations hold only over limited ranges, suggesting future work should explore more complex functions like piecewise linear or volcano-like functions reflecting the Sabatier principle in catalysis."

[7.3] Transfer learning from first-principles calculations to experiments with chemistry-informed domain transformation. Yuta Yahagi, Kiichi Obuchi, Fumihiko Kosaka, Kota Matsui. Machine Learning: Science and Technology (2025). https://doi.org/10.1088/2632-2153/adcdc0 Context: "The excerpt describes a transfer learning approach that integrates first-principles calculations with experimental chemistry. The method successfully transfers knowledge from computational models to experimental domains, achieving optimization in the target space with minimal experimental data (approximately 10 samples). The approach reduces computational model error by roughly an order of magnitude compared to models trained from scratch. The work explicitly integrates four scientific paradigms: experiment, theory, computation, and data. The authors propose this as a framework for theory-informed transfer learning between computation and experiment, suggesting it should be applicable to various materials and tasks. However, the excerpt does not provide specific details about particular domain pairs (e.g., catalysis to electrocatalysis) or the underlying mechanisms enabling knowledge portability across scientific domains."

[8.1] Optimizing cross-domain transfer for universal machine learning interatomic potentials. Jaesun Kim, Jinmu You, Yutack Park, Yunsung Lim, Yujin Kang, Jisu Kim, Haekwan Jeon, Suyeon Ju, Deokgi Hong, Seung Yul Lee, Saerom Choi, Yongdeok Kim, Jae W. Lee, Seungwu Han. Nature Communications (2026). https://doi.org/10.1038/s41467-026-70195-8 Context: "SevenNet-Omni achieves cross-domain transfer through a multi-task learning framework with two key strategies. First, selective regularization of task-specific parameters (θT) penalizes large values to encourage the model to rely on shared common representations across domains, suppressing overfitting to individual databases. Second, a domain-bridging set (DBS) augments training by evaluating subsets of one domain using computational methods from another domain, requiring minimal computational effort while substantially improving accuracy in out-of-distribution regions. The model was trained on 15 public databases covering molecular, crystalline, metal-organic framework, and surface systems using the SevenNet-MF architecture. A curriculum learning strategy was employed, first training on crystal databases to establish foundational knowledge, then progressively extending to molecular systems and the full diverse dataset. This approach enables effective knowledge transfer by maintaining accurate common potential energy surfaces across different chemical environments while strategically regularizing domain-specific contributions."

[8.2] Optimizing cross-domain transfer for universal machine learning interatomic potentials. Jaesun Kim, Jinmu You, Yutack Park, Yunsung Lim, Yujin Kang, Jisu Kim, Haekwan Jeon, Suyeon Ju, Deokgi Hong, Seung Yul Lee, Saerom Choi, Yongdeok Kim, Jae W. Lee, Seungwu Han. Nature Communications (2026). https://doi.org/10.1038/s41467-026-70195-8 Context: "SevenNet-Omni achieves cross-domain transfer through a multi-domain training strategy that jointly optimizes universal and task-specific parameters. The key innovations are: (1) a domain-bridging set (DBS) that aligns potential-energy surfaces across different datasets, using only small fractions (0.1%) of data; and (2) selective regularization that targets specific parameters to enhance out-of-distribution generalization while maintaining in-domain accuracy. The model was trained on 15 open datasets covering molecules, crystals, and surfaces. This approach enables state-of-the-art accuracy across diverse chemical domains, including adsorption-energy errors below 0.06 eV on metallic surfaces and 0.1 eV on metal-organic frameworks. Despite minimal training data from certain computational protocols (0.5% r2SCAN data), the model effectively transfers knowledge from larger datasets, demonstrating the scalability of bridging different quantum-mechanical fidelities and chemical domains."

[8.3] Optimizing cross-domain transfer for universal machine learning interatomic potentials. Jaesun Kim, Jinmu You, Yutack Park, Yunsung Lim, Yujin Kang, Jisu Kim, Haekwan Jeon, Suyeon Ju, Deokgi Hong, Seung Yul Lee, Saerom Choi, Yongdeok Kim, Jae W. Lee, Seungwu Han. Nature Communications (2026). https://doi.org/10.1038/s41467-026-70195-8 Context: "SevenNet-Omni achieves cross-domain transfer through a multi-task MLIP framework that divides model parameters into two types: shared parameters (θC) applied universally across all databases and task-specific parameters (θT) optimized exclusively for each database. Knowledge transfer occurs through the shared parameters, allowing learning from one domain to benefit others. The multi-domain training strategy optimizes cross-domain knowledge transfer by employing two key techniques: regularizing task-specific parameters and incorporating small cross-domain bridging sets. This approach addresses the challenge of training on heterogeneous datasets with different atomic configurations (inorganic versus organic systems) and computational approaches (different functionals). The framework treats each database as a distinct task to preserve intrinsic potential energy surface characteristics while enabling shared representation learning across domains. SevenNet-Omni was trained on 15 open databases containing 250 million structures across multiple chemical domains."

[8.4] Optimizing cross-domain transfer for universal machine learning interatomic potentials. Jaesun Kim, Jinmu You, Yutack Park, Yunsung Lim, Yujin Kang, Jisu Kim, Haekwan Jeon, Suyeon Ju, Deokgi Hong, Seung Yul Lee, Saerom Choi, Yongdeok Kim, Jae W. Lee, Seungwu Han. Nature Communications (2026). https://doi.org/10.1038/s41467-026-70195-8 Context: "SevenNet-Omni achieves cross-domain transfer through a curriculum learning strategy that progressively expands from inorganic bulk systems to organic molecules. The domain-bridging set (DBS) strategy involves subsampling approximately 125,000 structures (0.1%) from diverse databases totaling 125 million structures, with higher sampling weights assigned to structures structurally distant from inorganic systems and those computed with settings differing from PBE functional. For multi-domain training, selective regularization is achieved through oversampling certain databases to balance representation across inorganic crystals, catalytic surfaces, and molecular systems, with ratios determined by considering both structure count and average atoms per structure. The approach leverages effective transfer learning from abundant PBE-level data with similar potential energy surface characteristics, enabling performance gains compared to single-domain models."

[8.5] Optimizing cross-domain transfer for universal machine learning interatomic potentials. Jaesun Kim, Jinmu You, Yutack Park, Yunsung Lim, Yujin Kang, Jisu Kim, Haekwan Jeon, Suyeon Ju, Deokgi Hong, Seung Yul Lee, Saerom Choi, Yongdeok Kim, Jae W. Lee, Seungwu Han. Nature Communications (2026). https://doi.org/10.1038/s41467-026-70195-8 Context: "The excerpt discusses how SevenNet-Omni performs cross-domain transfer by training on multiple datasets spanning different material types and computational methods. The model demonstrates successful knowledge transfer across chemical domains (molecules and solids) and different functional fidelities (hybrid and semilocal functionals). A key aspect is selective task regularization, which enables 7net-Omni to maintain more uniform error levels across different domains compared to other multi-task models. The approach evaluates performance using small-molecule torsion barriers as benchmarks for conformational potential energy surface (PES) descriptions, with models reoptimized at different functional levels (PBE-D3, ωB97M-D3) to directly compare energy barriers. The multi-task training framework requires effective knowledge transfer from PESs learned with hybrid functionals to those following PBE functionals for molecules and solid-state systems simultaneously."

[9.1] The Open Catalyst 2022 (OC22) Dataset and Challenges for Oxide Electrocatalysts. Richard Tran, Janice Lan, Muhammed Shuaibi, Brandon M. Wood, Siddharth Goyal, Abhishek Das, Javier Heras-Domingo, Adeesh Kolluru, Ammar Rizvi, Nima Shoghi, Anuroop Sriram, Félix Therrien, Jehad Abed, Oleksandr Voznyy, Edward H. Sargent, Zachary Ulissi, C. Lawrence Zitnick. ACS Catalysis (2023). https://doi.org/10.1021/acscatal.2c05426 Context: "The excerpt presents key transfer learning results between

OC20 (metals dataset) and OC22 (oxides dataset). When combining these chemically dissimilar datasets, GemNet-OC achieved approximately 36% improvement in energy predictions through fine-tuning. Additionally, joint training of both datasets yielded 19% improvement in total energy predictions on OC20 and 9% improvement in force predictions on OC22. The OC20 dataset contained 250 million single-point calculations focused on metal catalysts with various adsorbates on metal surfaces, while OC22 was specifically developed to address the lack of oxide materials in training data. This demonstrates that cross-domain training between metals and oxides provides measurable benefits for model generalization across different material systems."

[9.2] The Open Catalyst 2022 (OC22) Dataset and Challenges for Oxide Electrocatalysts. Richard Tran, Janice Lan, Muhammed Shuaibi, Brandon M. Wood, Siddharth Goyal, Abhishek Das, Javier Heras-Domingo, Adeesh Kolluru, Ammar Rizvi, Nima Shoghi, Anuroop Sriram, Félix Therrien, Jehad Abed, Oleksandr Voznyy, Edward H. Sargent, Zachary Ulissi, C. Lawrence Zitnick. ACS Catalysis (2023). https://doi.org/10.1021/acscatal.2c05426 Context: "The excerpt describes experimental approaches for leveraging OC20 and OC22 datasets together for catalyst discovery. Three training strategies are explored: OC22-only (baseline), joint training (combining OC20+OC22 data), and fine-tuning (initializing with OC20 pretrained weights, then training on OC22). For S2EF-Total tasks, joint training experiments used different proportions of OC20 data (2M, 20M, and All). The authors note that while comprehensive datasets spanning all applications and chemical diversity are computationally infeasible, leveraging large existing datasets like OC20 can enable training effective models. The excerpt indicates these experiments aim to provide insights beyond just OC22 performance, though specific findings on cross-domain transfer between metals and oxides or out-of-distribution generalization results are not detailed in this section."

[9.3] The Open Catalyst 2022 (OC22) Dataset and Challenges for Oxide Electrocatalysts. Richard Tran, Janice Lan, Muhammed Shuaibi, Brandon M. Wood, Siddharth Goyal, Abhishek Das, Javier Heras-Domingo, Adeesh Kolluru, Ammar Rizvi, Nima Shoghi, Anuroop Sriram, Félix Therrien, Jehad Abed, Oleksandr Voznyy, Edward H. Sargent, Zachary Ulissi, C. Lawrence Zitnick. ACS Catalysis (2023). https://doi.org/10.1021/acscatal.2c05426 Context: "The excerpt describes findings from joint training on OC20 and OC22 datasets for catalyst discovery. Key findings include: (1) Naively fitting models on both datasets—despite OC22 being much smaller—surprisingly improved accuracy for predicting OC20 energies; (2) Models trained on both datasets simultaneously achieved the highest accuracy for both OC20 and OC22; (3) Both dataset combinations followed the same log-log scaling for energy mean absolute error; (4) Training on both datasets enabled learning of common representations shared across both datasets. However, the excerpt does not specifically address cross-domain training between metals and oxides or out-of-distribution generalization for adsorption energies. The text suggests these multi-dataset approaches could extend to other large datasets like NOMAD and Materials Project."

[10.1] Transfer Learning for Deep Learning-based Prediction of Lattice Thermal Conductivity. L. Klochko, M. d'Aquin, A. Togo, L. Chaput. ArXiv (2411). https://doi.org/10.48550/arxiv.2411.18259 Context: "The excerpt references several major public materials databases relevant for machine learning applications. These include: (1) AFLOW - a standard for high-throughput materials science calculations; (2) OQMD (Open Quantum Materials Database) - assessing DFT formation energies; (3) Materials Project - described as a materials genome approach to accelerating materials innovation; and (4) JARVIS (Joint Automated Repository for Various Integrated Simulations) - for data-driven materials design. These databases are cited in the context of a paper on transfer learning for deep learning-based prediction of lattice thermal conductivity. The excerpt does not provide details about specific material properties available in each database or their suitability for cross-domain transfer learning pairs."

[11.1] Transfer learning on large datasets for the accurate prediction of material properties. Noah Hoffmann, Jonathan Schmidt, S. Botti, Miguel A. L. Marques. JournalArticle (2023). https://doi.org/10.5281/zenodo.8143755 Context: "The excerpt contains a reference list that mentions several major public materials databases and datasets relevant for materials property prediction. The Open Quantum Materials Database (OQMD) is referenced as a resource for assessing material properties. The Materials Project is mentioned as a materials genome approach to accelerating materials innovation. AFLOW is described as an

automatic framework for high-throughput materials discovery. The Open Catalyst 2020 (OC20) dataset is cited as a community challenge dataset. Additionally, the excerpt references work on crystal graph convolutional neural networks and various transfer learning approaches for materials property prediction, indicating these databases are used in conjunction with machine learning methods. However, the excerpt does not provide detailed information about specific properties (band gap, thermal conductivity, catalytic activity) stored in these databases or how they function as donor-recipient pairs for cross-domain transfer learning."

[12.1] Developing machine learning for heterogeneous catalysis with experimental and computational data.. Carlota Bozal-Ginesta, Sergio Pablo-García, Changhyeok Choi, A. Tarancón, Alán Aspuru-Guzik. Nature reviews. Chemistry (2025). https://doi.org/10.1038/s41570-025-00740-4 Context: "The excerpt references several major public databases for materials science and catalysis data that could support machine learning applications. These include: Catalysis-Hub.org for catalysis-related data; NOMAD, a distributed web-based platform for materials science research data; AFLOW (aflow.org), described as a web ecosystem of databases, software and tools; the Open Quantum Materials Database (OQMD) featuring high-throughput density functional theory calculations; the Materials Project, a materials genome approach; and the Open Catalyst datasets (OC20 from 2021 and OC22 from 2023) focusing on oxide electrocatalysts. Additionally, ioChem-BD platform is mentioned for managing computational chemistry data. These repositories provide computational and experimental data that could potentially serve as sources for transfer learning applications across different materials properties and domains."

[13.1] AQCat25: unlocking spin-aware, high-fidelity machine learning potentials for heterogeneous catalysis. Omar Allam, Brook Wander, SungYeon Kim, Rudi Plesch, Tyler Sours, Jia-Min Chu, Thomas Ludwig, Jiyoon Kim, Rodrigo Wang, Shivang Agarwal, Alan Rask, Alexandre Fleury, Chuhong Wang, Andrew Wildman, Thomas Mustard, Kevin Ryczko, Paul Abruzzo, AJ Nish, Aayush R. Singh. npj Computational Materials (2026). https://doi.org/10.1038/s41524-026-02099-6 Context: "The excerpt contains references to foundational work on linear scaling relationships and Brønsted-Evans-Polanyi (BEP) relations in heterogeneous catalysis. Nørskov et al. (2002) established the concept of 'Universality in heterogeneous catalysis,' indicating these relationships provide generalizable principles across catalyst systems. Bligaard et al. (2004) specifically examined 'The Brønsted–Evans–Polanyi relation and the volcano curve in heterogeneous catalysis,' which are key frameworks for relating activation energies to free energies of reaction intermediates. Van Santen et al. (2009) developed 'Reactivity theory of transition-metal surfaces' using linear activation energy-free energy analysis. These references suggest these relations enable scaling of catalytic properties across different transition metal catalysts. However, the excerpt does not provide detailed mechanistic explanations of how knowledge transfer occurs, when these relations break down, or discuss alternative approaches for cases where they fail."

[14.1] A Systematic Survey and Benchmark of Deep Learning for Molecular Property Prediction in the Foundation Model Era. Zongru Li, Xingsheng Chen, Honggang Wen, Regina Qianru Zhang, Ming Li, Xiaojin Zhang, Hongzhi Yin, Qiang Yang, Kwok-Yan Lam, Pietro Lio, Siu-Ming Yiu. Journal of Chemical Theory and Computation (2026). https://doi.org/10.1021/acs.jctc.5c02081 Context: "The excerpt demonstrates several mechanisms of knowledge transfer across scientific domains. Chemical foundation models pretrained on extensive molecular libraries create versatile representations enabling few-shot prediction of diverse pharmacological properties with minimal domain-specific data. Transfer learning enables models pretrained on pharmacological datasets to predict environmental properties like aquatic toxicity and biodegradability, crucial for PFAS screening where conventional models struggle. Multitask GNNs leverage shared substructural features across related prediction tasks. The foundation model strategy of pretraining generic potentials followed by fine-tuning unifies static property prediction with dynamical behavior. Graph-based models demonstrate significant out-of-distribution generalization, with GNoME identifying experimentally confirmed stable crystal structures far beyond training distributions. These mechanisms rely on learning generalizable molecular representations that capture fundamental chemical patterns transferable across domains, reducing data requirements while maintaining accuracy."

[15.1] Multi-fidelity transfer learning for quantum chemical data using a robust density functional tight binding baseline. Mengnan Cui, Karsten Reuter, Johannes T. Margraf. Machine Learning: Science and Technology (2025). https://doi.org/10.1088/2632-2153/adc222 Context: "The excerpt identifies that negative transfer in multi-fidelity transfer learning for chemical data can be driven by noise from low-fidelity methods, specifically mentioning density functional tight binding as a baseline source of noise that significantly impacts fine-tuned models. The paper also notes that limited overlap between datasets for pre-training and fine-tuning is a key factor causing transfer learning to fail, as models cannot generalize effectively when there are insufficient shared features or representations. The authors emphasize that the overlap should be considered in terms of both the configuration space covered by structures and the fidelities of quantum chemical methods. However, the excerpt does not provide specific quantitative results about when pretraining hurts out-of-distribution performance or detailed conditions under which negative transfer occurs in chemistry and materials science ML beyond these general observations."